POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE | Full Marks : 100
Course: Computer Networks (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define Computer Network. Trace the evolution of telecommunication technologies from 1G to 5G, highlighting the key advancements in each generation. 7
   b) List out the devices used for inter-networking. Compare TCP/IP model with OSI reference model. 8
2. a) Explain Circuit Switching, Message switching and Packet switching in detail. Why is Packet Switching the dominant method for the modern Internet. 7
   b) Explain different types of guided and unguided media. Which media is considered the most effective for data transmission? Why? 8
3. a) Explain sliding window flow control mechanism in detail. How is Go Back N ARQ different from Selective Reject ARQ? 7
   b) Illustrate checksum method for error detection with an example. 8

OR

How is slotted ALOHA different from Pure ALOHA? Explain the algorithm of CSMA/CD and draw the flowchart.

4. a) Given an IP address of 192.168.2.0 / 25, you are responsible to create three subnetworks for the departments namely Marketing, Sales, and Accounts that require 35, 11, and 5 hosts respectively. Find out the subnetwork address, broadcast address, and ranges of host addresses, subnetwork masks and wildcard masks for each department. 7
   b) Is it possible to use both IPv4 and IPv6 simultaneously? If yes, how? Compare the header format of Internet Protocol version 4 (IPv4) with IPv6 and explain each fields in short. 8

OR

How does a router learn and update its routing table? Explain the process with suitable routing algorithm.

5. a) List out and explain TCP client/server socket flow with suitable diagram. 7
   b) Discuss working principles of traffic shaping algorithms (Leaky Bucket, Token Bucket). 8
6. a) What are the services provided by Application Layer. Explain SMTP, POP and IMAP, HTTP. 8
   b) What is cryptography? How Diffie-Hellman algorithm negotiate a shared key between sender and receiver. Explain with example. 7
7. Write short notes on: **(Any two)** 2×5
   a) Firewall and its types
   b) NAT
   c) DHCP Principles

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